

TRACKING THE VIRAL THREAT TO UZBEKISTAN’S HONEY BEES IN SPRING 2024

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Abstract. *The health of honey bees, a cornerstone of the beekeeping industry, is under threat from various viruses that can cause symptoms ranging from wing deformities to paralysis, thereby reducing productivity. This study focuses on the detection and analysis of seven major honey bee viruses in Uzbekistan, a critical region on the Silk Road that could play a significant role in the transmission of pathogens due to its strategic location and active exchange with neighboring countries. Utilizing multiplex one-step RT-PCR, we conducted diagnostics on samples collected from 20 sites across four regions in Uzbekistan during the spring of 2024. Our findings reveal the presence of black queen cell virus in all samples and deformed wing virus in nearly all, with regional variations in virus occurrence. Notably, chronic bee paralysis virus and sacbrood virus were detected in specific regions, indicating potential localized spread. Sequence analysis showed high identity with previous isolates from Uzbekistan, suggesting regional transmission patterns. These results highlight the urgent need for comprehensive research on honey bee viruses in Uzbekistan and underscore the importance of monitoring these pathogens to safeguard the apiculture industry and biodiversity.*

Key words: *detection, honey bee, RT-PCR, Uzbekistan, virus.*

In the beekeeping industry, the health of honey bees is a critical factor, and viruses are a significant pathogen impacting their well-being (Gisder and Genersch, 2015). Infected honey bees often exhibit symptoms such as wing deformities and paralysis, which directly reduce their productivity (de Miranda et al., 2010; de Miranda and Genersch, 2010; Genersch and Aubert, 2010). However, in some cases, honey bees show no external signs of infection, leading to the detection of multiple viral infections in seemingly healthy colonies upon diagnosis (Amiri et al., 2020). Reports on honey bee viruses are increasingly being confirmed with the revolutionary advancements in diagnostic technologies (Beaurepaire et al., 2020). As with other viruses, molecular diagnostic methods based on nucleic acid amplification, such as (RT-)PCR, and serological diagnostic methods based on the interaction between viral proteins and their corresponding antibodies, are primarily used (Cassedy et al., 2021; Cuc et al., 2021). Virome analysis based on Next-Generation Sequencing (NGS) technology has made it easier to identify the presence of various known and novel viruses within individual honey bees, leading to the continual reporting of novel viruses such as the Lake Sinai virus (Kwon et al., 2023; Runckel et al., 2011). Furthermore, the ease of virus detection has facilitated the observation of international movement patterns of various viruses, confirming that these pathogens are not confined to specific regions but are emerging across continents. The spread of these viruses is presumed to be driven

by the movement of infected honey bee colonies or by vectors such as mites that can transmit several key viruses (McMenamin and Genersch, 2015).

Situated at the heart of the Silk Road, bridging Asia and Europe, Uzbekistan could serve as a pivotal intermediary in the transmission of pathogens. Additionally, the active exchange with Russia and Asian countries further amplifies the potential for virus mediation. Despite this significance, research on honey bee pathogens in Uzbekistan has been scant. The significance of researching honey bee virus patterns in Uzbekistan is not only vital for the country’s apiculture industry but also for understanding the spread of viruses in neighboring nations and beyond, into Europe and Asia.

We conducted diagnostics for seven major species of viruses [acute bee paralysis virus (ABPV), black queen cell virus (BQCV), chronic bee paralysis virus (CBPV), deformed wing virus (DWV), Israeli acute paralysis virus (IAPV), Kashmir bee virus (KBV) and sacbrood virus (SBV)] in honey bee (*Apis mellifera*) samples collected in the spring of 2024 from 20 sites of four regions in Uzbekistan: Samarqand, Jizzax, Navoiy, and Qashqadaryo (**Table 1**).

Table 1. Sample information

No.	Collection (label)	site Number repetitions	of Circumstance	Coordinate
1	Jizzax (Ji)	5	Agricultural area	40°05'03.0"N 67°48'26.0"E
2			Agricultural area	40°04'51.0"N 67°53'26.0"E
3			Urban area	40°08'26.0"N 67°47'46.0"E
4			Urban area	40°08'47.0"N 67°45'16.0"E
5			Urban area	40°08'43.0"N 67°45'21.0"E
6	Navoiy (Na)	4	Urban area	40°06'36.2"N 65°20'57.3"E
7			Urban area	40°07'42.0"N 65°22'13.9"E
8			Agricultural area	40°01'48.8"N 64°48'16.3"E
9			Urban area	40°04'51.4"N 65°22'08.9"E
10	Samarqand (Sa)	5	Urban area	39°37'43.0"N 67°02'32.0"E
11			Urban area	39°37'44.0"N 67°02'32.0"E
12			Urban area	39°37'45.0"N 67°02'32.0"E
13			Agricultural area	39°28'34.0"N 67°02'24.0"E
14			Agricultural area	39°25'28.0"N 67°02'20.0"E
15	Qashqadaryo (Ka)	6	Agricultural area	38°57'18.0"N 66°45'10.0"E
16			Agricultural area	38°57'34.0"N 66°45'19.0"E
17			Agricultural area	38°59'55.0"N 66°38'54.0"E
18			Agricultural area	38°59'55.0"N 66°38'38.0"E
19			Agricultural area	38°59'31.0"N 66°39'12.0"E
20			Agricultural area	38°59'15.0"N 66°39'13.0"E

Pooling was performed by randomly selecting two honey bees from each of the five hives sampled from each farm, resulting in a total of ten bees per pooling group, followed by the extraction of total RNA using the TRIzol® protocol. The quality of the extracted RNA was measured with a NanoPhotometer® NP80 (IMPLEN, Munich, Germany). Virus diagnostics from the prepared RNA were carried out via multiplex one-step RT-PCR, utilizing SuPrimeScript

RT-PCR Premix (2X) (GENETBIO, Daejeon, South Korea), employing eight specific primers for seven viruses as introduced in **Table 2**. During multiplex RT-PCR, the eight primers were grouped into three based on the length of the amplification products, and the results were verified through agarose gel electrophoresis.

Table 2. Primer sets used in this study

Virus	Sequence (5' → 3')	T _m (°C)	Size (bp)
ABPV	TGAGAACACCTGTAATGTGGGT GGCAGGTATAGTATTTTCGGAACG	56	1271
BQCV	AAATGCCAATGTGGACCAAA GATAGGGCTGCTATCCACCG	56	370
CBPV RNA1	CCTCCCGTCATGATTTCCCC TGCATTGTCATTGCTGGAAC	56	974
CBPV RNA2	CGTTATCTCAGGTCCGGCAA GAGCAAAGACGCTGAGGAGA	56	612
DWV	CGGTGCGACTGAACTTCTA CATACGTTCTTGCTCCAGCG	56	610
IAPV	ATTCCTGTGTCGGAGCAGTG CAAAGTATCCTCAAGTTGTGGG	56	870
KBV	GATGAACGTCGACCATTGA TGTGGGTGGCTATGAGTCA	56	414
SBV	GGAGGCCTGGGAAAAGAGTG TTCCAACCTGCACCACAGGTT	56	535

Amplification products identified were subjected to Sanger sequencing (Macrogen, Seoul, South Korea) for base sequence verification, and the outcomes were analyzed using NCBI BLAST and CLC genomics workbench (version 24.0.1).

According to the RT-PCR results, ABPV, IAPV, and KBV were not detected in samples from any region, whereas BQCV was found in all samples. DWV was also detected in almost all samples, except for two. CBPV was only identified in the Jizzax and Navoiy regions, and SBV was found exclusively at two sites in the Navoiy region (**Table 3**). Although further analysis on more samples is required, these results indicate regional variations in the occurrence of viruses.

Table 3. Results of RT-PCR diagnostics for seven virus species at 20 sites secured in this study

Virus	Jizzax					Navoiy				Samarqand					Qashqadaryo					
	1	2	3	4	5	1	2	3	4	1	2	3	4	5	1	2	3	4	5	6
ABPV	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
BQCV	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
CBPV RNA1	D	D	D	D	N	D	D	D	D	N	N	N	N	N	N	N	N	N	N	N
CBPV RNA2	D	D	D	D	N	D	D	D	D	N	N	N	N	N	N	N	N	N	N	N
DWV	D	D	D	D	D	D	D	D	D	D	D	D	D	N	D	D	N	D	D	D
IAPV	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
KBV	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
SBV	N	N	N	N	N	N	D	D	N	N	N	N	N	N	N	N	N	N	N	N

* ‘D’ indicates a positive result, and ‘N’ indicates a negative result for each virus

Although only partial sequences were analyzed, when the PCR amplification products of each virus were examined through Sanger sequencing, it was confirmed that they showed the highest identity with the sequences identified by our team in Uzbekistan in 2022. DWV exhibited a 98% identity with the isolates from Yanghikurgan (OR912391.1) and Galvasay (OR912375.1), and BQCV showed around 98% similarity with the Yanghikurgan isolates (OR912388.1 and OR912392.1) (**Table 4**). CBPV, which had not been detected by this research team previously, was identified for the first time in Uzbekistan; for RNA1, sample 1 showed 94.24% similarity with an isolate collected from honey bees in Chongqing, China, in 2019 (MZ821995.1), and sample 2 was 96.04% similar to an isolate separated in Slovenia in 2010 (KY937971.1). RNA2 was 98% similar to an isolate separated from honey bees in Austria in 2018 (MK637523.1). While there may be some differences when analyzing the complete genome sequence, the regional characteristics bridging European and Asian countries were partially reflected in the sequence similarities.

Table 4. BLAST results for the amplicons in this study

Virus	Sample Name	Accession	Isolate	Max Score	Total Score	Query Coverage	E value	Per. Ident
DWV	Ji2	OR912375.1	Galvasay (U1)	1051	1051	99%	0	98.33%
		OR912391.1	Yanghikurgan (U4)	1046	1046	99%	0	98.16%
		KY909333.1	Vespa_crabro_DWV_PI_2016	1024	1024	99%	0	97.50%
		OR361537.1	DWV-A-OR-OCT2018-EVR24	1024	1024	99%	0	97.50%
		MH267696.1	DWV_MS	1013	1013	99%	0	97.16%
	Sa2	OR912391.1	Yanghikurgan (U4)	985	985	100%	0	98.06%
		OR912375.1	Galvasay (U1)	985	985	100%	0	98.06%
		KY909333.1	Vespa_crabro_DWV_PI_2016	957	957	100%	0	97.18%
		OR361537.1	DWV-A-OR-OCT2018-EVR24	957	957	100%	0	97.18%
		OR496436.1	Bonghwa	952	952	100%	0	97.00%
BQCV	Na1	OR912392.1	Yanghikurgan (U4)	536	536	99%	1.00E-147	98.05%
		OR912388.1	Yanghikurgan (U3)	531	531	99%	6.00E-146	97.73%
		OR496415.1	Uiseong	508	508	99%	3.00E-139	96.43%
		MZ821800.1	BQCV_No1_Acc003-BJ2019	508	508	99%	3.00E-139	96.43%

CBP V RNA 1	Qa1	MZ821804. 1	BQCV_No5_Acc019- SD2017	505	505	99%	4.00E -138	96.10 %
		OR912392. 1	Yanghikurgan (U4)	224	224	100%	4.00E -54	98.44 %
		OR912388. 1	Yanghikurgan (U3)	224	224	100%	4.00E -54	98.44 %
		KY741959. 1	GS1	207	207	100%	4.00E -49	96.09 %
		OR496405. 1	Andong	207	207	100%	4.00E -49	96.09 %
		OR496415. 1	Uiseong	206	206	99%	1.00E -48	96.06 %
	Na1	MZ821995. 1	CBPV_No22_Am066- CQ2019	1221	1221	99%	0	94.24 %
		MZ821969. 1	CBPV_No9_Am010- GS2018	1218	1218	99%	0	94.12 %
		MZ821963. 1	CBPV_No6_AM-AH-2017- 3	1201	1201	99%	0	93.74 %
		MZ821979. 1	CBPV_No14_Am025- HLJ2017	1199	1199	99%	0	93.73 %
		OK491520. 1	known_11	1199	1199	99%	0	93.74 %
		KY937971. 1	M92/2010	1439	1439	99%	0	96.04 %
		ON648749. 1	CBPV RNA1 376/2020	1428	1428	99%	0	95.81 %
		OR582946. 1	FR20HM439	1406	1406	99%	0	95.36 %
		OR582954. 1	FR20PA427.1	1400	1400	99%	0	95.25 %
		OR582951. 1	FR19HM425.0'	1397	1397	99%	0	95.02 %
	Na1	MK637523. 1	AUT-17 segment RNA2, complete sequence	1066	1066	99%	0	98.83 %
		KY937972. 1	M92/2010	1055	1055	99%	0	98.49 %
		ON648750. 1	CBPV RNA2 376/2020	1055	1055	99%	0	98.49 %
		ON648752. 1	CBPV RNA2 341/2019	1050	1050	99%	0	98.33 %
		MZ821976. 1	CBPV_No12_Am020- HB2019	1005	1005	99%	0	96.99 %
		MK637523. 1	AUT-17	1022	1022	100%	0	98.61 %
		KY937972. 1	M92/2010	1011	1011	100%	0	98.27 %
		ON648750. 1	CBPV RNA2 376/2020	1011	1011	100%	0	98.27 %

ON648752. 1	CBPV RNA2 341/2019	1005	1005	100%	0	98.09 %
MZ821976. 1	CBPV_No12_Am020- HB2019	944	944	100%	0	96.19 %

The recent findings underscore the significance of research on honey bee pathogens, which has been inadequately conducted in Uzbekistan and neighboring countries, despite the importance of the apiculture industry. Building upon these results, further research—including virome analysis to obtain the full-length genome nucleotide sequences of the viruses identified in this study, and to check for the emergence of novel or mutant strains—could be a valuable asset in tracking the spread of honey bee viruses.

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